EMCH 792: Optimal State Estimation
Homework 3
Written Portion Solutions

Instructor: N. Vitzilaios, Department of Mechanical Engineering
University of South Carolina
Solutions by: J.C. Vaught

Due: February 26, 2026

Table of Contents

Exercise 2.5 (Page 75): Exponentially Distributed Random Variable

Given

$$f_X(x) = \begin{cases} \alpha e^{-\alpha x}, & x > 0, \\ 0, & x \le 0, \end{cases} \qquad \alpha > 0.$$

Derivation

(a) Probability distribution function (CDF)

$$F_X(x) = P(X \le x) = \begin{cases} 0, & x \le 0, \\ \int_0^x \alpha e^{-\alpha z}\, dz = 1 - e^{-\alpha x}, & x > 0. \end{cases}$$

(b) Mean

$$\mu_X = E[X] = \int_0^\infty x\alpha e^{-\alpha x} dx = \frac{1}{\alpha}.$$

(c) Second moment

$$E[X^2] = \int_0^\infty x^2 \alpha e^{-\alpha x} dx = \frac{2}{\alpha^2}.$$

(d) Variance

$$\sigma_X^2 = E[X^2] - \mu_X^2 = \frac{2}{\alpha^2} - \frac{1}{\alpha^2} = \frac{1}{\alpha^2}.$$

So $\sigma_X = 1/\alpha$.

(e) Probability within one standard deviation of the mean

$$P(\mu_X - \sigma_X \le X \le \mu_X + \sigma_X) = P\left(0 \le X \le \frac{2}{\alpha}\right) = F_X\left(\frac{2}{\alpha}\right) - F_X(0) = 1 - e^{-2}.$$

$$1 - e^{-2} \approx 0.8646647.$$

Final Result

$$F_X(x) = \begin{cases} 0, & x \le 0 \\ 1 - e^{-\alpha x}, & x > 0 \end{cases}, \quad E[X] = \frac{1}{\alpha}, \quad E[X^2] = \frac{2}{\alpha^2}, \quad \mathrm{Var}(X) = \frac{1}{\alpha^2},$$

$$P\left(|X - \mu_X| \le \sigma_X\right) = 1 - e^{-2} \approx 0.8647.$$

Exercise 2.8 (Page 76): Sum of Two Zero-Mean Uncorrelated RVs

Let W and V be zero-mean, uncorrelated RVs with standard deviations σ_W and σ_V. Define

$$X = W + V.$$

Derivation

Since $E[W] = E[V] = 0$, then $E[X] = 0$.
Using uncorrelatedness, $\mathrm{Cov}(W, V) = 0$. Therefore,

$$\mathrm{Var}(X) = \mathrm{Var}(W + V) = \mathrm{Var}(W) + \mathrm{Var}(V) + 2\mathrm{Cov}(W, V) = \sigma_W^2 + \sigma_V^2.$$

Hence,

$$\sigma_X = \sqrt{\sigma_W^2 + \sigma_V^2}.$$

Final Result

$$\boxed{\sigma_X = \sqrt{\sigma_W^2 + \sigma_V^2}}$$

Exercise 2.12 (Page 76): $Y(t) = X \cos t$

Suppose X is a random variable and

$$Y(t) = X \cos t.$$

Derivation

(a) Expected value

$$E[Y(t)] = E[X \cos t] = (E[X]) \cos t = \bar{x} \cos t.$$

(b) Time average

$$A[Y(t)] = \lim_{T\to\infty} \frac{1}{T} \int_0^T X \cos t \, dt = X \lim_{T\to\infty} \frac{\sin T}{T} = 0.$$

(c) Condition for equality For the ensemble mean to match the time average for all t,

$$\bar{x} \cos t = 0 \quad \forall t \implies \bar{x} = 0.$$

Also, for an individual realization $y(t) = x \cos t$ to equal its time average for all t, one needs $x = 0$.

Final Result

$$E[Y(t)] = \bar{x} \cos t, \qquad A[Y(t)] = 0.$$

Mean/time-average equality for all t requires zero mean ($\bar{x} = 0$). For a specific realization to be identically equal to its time average, $x = 0$.